SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	23-07-2026	**Duration:**	60 minutes

Code of Conduct

I A. Govardhan Reddy (192421359) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Govardhan Reddy

Project:

SkillSwap – A Peer-to-Peer Local Skill-Sharing & Micro-Learning Platform

Problem Statement

Across most cities and campuses, people possess valuable skills — cooking, coding, music, languages, fitness, crafts — that they would happily teach informally, and at the same time want to learn new skills from others nearby. However, there is no simple, trustworthy way to discover local peers with complementary skills, agree on a fair exchange, schedule a session, and build a reputation over time. Existing options are polarized: formal e-learning platforms are expensive and impersonal, while generic classifieds/marketplace apps are not designed for skill bartering, lack scheduling or trust mechanisms, and are unsafe for in-person meetups.

SkillSwap addresses this gap by providing a peer-to-peer platform where users list the skills they can teach and the skills they wish to learn, get matched with nearby or virtually-available peers, schedule sessions, exchange skills using a fair "skill-credit" system (with optional cash top-ups), and build trust through verified profiles, ratings and reviews.

1. Project Vision and Stakeholders

1.1 Project Vision

To build a trusted, community-driven platform that empowers anyone to teach and learn skills locally through intelligent peer matching, flexible scheduling, and a fair skill-credit exchange economy — turning everyday knowledge into accessible, lifelong learning opportunities and stronger local communities.

1.2 Project Objectives

- Enable users to create profiles listing skills they can teach and skills they want to learn.
- Match learners and teachers based on skill compatibility, location/proximity, availability, and ratings.
- Provide an integrated scheduling and booking system for in-person or virtual sessions.
- Implement a skill-credit economy so users can exchange skills fairly without mandatory cash payment.
- Establish trust and safety through identity verification, ratings, reviews, and reporting mechanisms.
- Foster community growth through skill groups, local events, and notifications.
- Deliver a responsive, easy-to-use mobile and web experience for a non-technical, diverse user base.

1.3 Key Stakeholders

Stakeholder	Role / Interest
Product Owner	Owns the product vision, prioritizes the backlog, represents business and customer interests, accepts/rejects completed work.
Scrum Master	Facilitates Scrum ceremonies, removes impediments, coaches the team on Agile practices and protects the sprint.
Development Team (Frontend, Backend, Mobile)	Designs, builds, and tests the platform features; estimates and commits to sprint backlog items.
UI/UX Designer	Designs intuitive onboarding, matching, and booking flows; ensures accessibility and usability.
QA / Test Engineers	Validates acceptance criteria, performs functional, regression, and usability testing.
End Users – Teachers	Individuals offering skills; interested in visibility, fair credit exchange, and safe scheduling.
End Users – Learners	Individuals seeking to learn skills; interested in easy discovery, trust signals, and flexible booking.
Community Moderators	Monitor content, verify profiles, handle disputes/reports, and maintain platform safety.
Business Sponsor / Investor	Funds the project; interested in user growth, retention, and monetization (credit top-ups, premium features).
Payment & Compliance Officer	Ensures the skill-credit and payment system complies with financial and data-privacy regulations.
Marketing Team	Drives user acquisition and community campaigns; provides feedback on positioning and messaging.

2. Epics and User Stories

The product requirements have been decomposed into six Epics. Each Epic is broken down into well-structured User Stories following the standard Agile format: "As a <user>, I want <feature>, so that <benefit>." Story IDs follow the pattern US-E<epic#>-<seq>.

Epic 1 (E1): User Onboarding & Profile Management

ID	User Story	Priority
US-E1-01	As a new user, I want to sign up using email, phone, or social login, so that I can create an account quickly and securely.	High
US-E1-02	As a user, I want to create a profile listing skills I can teach and skills I want to learn, so that others can discover me for matching.	High
US-E1-03	As a user, I want to verify my identity (email/phone/ID), so that other users can trust my profile.	High
US-E1-04	As a user, I want to set my location and availability, so that I get matched with realistic, nearby, or timezone-compatible peers.	Medium
US-E1-05	As a user, I want to edit or deactivate my profile at any time, so that I control my data and visibility.	Low

Epic 2 (E2): Skill Listing & Discovery

ID	User Story	Priority
US-E2-01	As a teacher, I want to list a skill with description, level, and format (in-person/virtual), so that learners understand what I offer.	High
US-E2-02	As a learner, I want to search and filter skills by category, location, and rating, so that I can find a suitable teacher quickly.	High
US-E2-03	As a learner, I want to view a teacher's profile, ratings, and reviews before booking, so that I can make an informed decision.	High
US-E2-04	As a user, I want to receive personalized skill recommendations, so that I discover relevant learning opportunities without searching manually.	Medium

Epic 3 (E3): Matching & Scheduling

ID	User Story	Priority
US-E3-01	As a learner, I want to request a session with a matched teacher, so that I can start learning a skill.	High
US-E3-02	As a teacher, I want to accept, decline, or propose alternate times for a session request, so that I can manage my schedule.	High
US-E3-03	As a user, I want to view a calendar of my upcoming sessions, so that I never miss a scheduled exchange.	Medium
US-E3-04	As a user, I want to receive reminders before a scheduled session, so that I don't forget or arrive late.	Medium
US-E3-05	As a user, I want to reschedule or cancel a session within policy limits, so that I can handle changes in my availability.	Low

Epic 4 (E4): Skill-Credit Exchange & Payments

ID	User Story	Priority
US-E4-01	As a user, I want to earn skill-credits when I teach a session, so that I can redeem them to learn from others.	High
US-E4-02	As a user, I want to spend skill-credits (or top up with cash) to book a session, so that I can learn even before I have taught.	High
US-E4-03	As a user, I want to view my skill-credit balance and transaction history, so that I can track my exchanges.	Medium
US-E4-04	As a user, I want a secure payment gateway for optional cash top-ups, so that my financial information stays protected.	Medium

Epic 5 (E5): Trust, Safety & Reviews

ID	User Story	Priority
US-E5-01	As a learner, I want to rate and review a teacher after a session, so that I can share feedback and help other learners.	High
US-E5-02	As a user, I want to report inappropriate behavior or profiles, so that the platform stays safe for the community.	High
US-E5-03	As a user, I want to see verification badges on profiles, so that I can quickly assess trustworthiness.	Medium
US-E5-04	As a moderator, I want a dashboard to review reported content, so that I can take timely action on violations.	Medium

Epic 6 (E6): Community & Notifications

ID	User Story	Priority
US-E6-01	As a user, I want to receive push/email notifications for session requests and updates, so that I stay informed in real time.	Medium
US-E6-02	As a user, I want to join local or interest-based skill groups, so that I can connect with a broader community.	Low
US-E6-03	As a user, I want to see upcoming community skill-swap events near me, so that I can participate in group learning.	Low

3. Acceptance Criteria

Acceptance criteria have been defined for the highest-priority user stories using the Given–When–Then format, ensuring each story is specific, measurable, and testable before development begins.

US-E1-02 — Create profile with skills to teach/learn

Given	When	Then
the user is logged in and on the Profile Setup screen	the user adds at least one "skill I can teach" and one "skill I want to learn" with a description	the profile is saved and displayed on the user's public profile with both skill lists visible
a user attempts to save a profile	no skill has been added to either list	the system displays a validation message and prevents saving until at least one skill is added

US-E1-03 — Identity verification

Given	When	Then
a registered user submits a valid phone number or ID document for verification	the verification service confirms the identity	a "Verified" badge appears on the user's profile within 24 hours
a user's verification document is rejected	the review is completed	the user is notified with a reason and given an option to resubmit

US-E2-02 — Search and filter skills

Given	When	Then
a learner is on the Discover Skills screen	the learner applies filters for category, distance, and minimum rating	the results list updates to show only teachers matching all selected filters, sorted by relevance
no teacher matches the applied filters	the search executes	the system displays a "no matches found" message with a suggestion to broaden filters

US-E3-01 — Request a session

Given	When	Then
a learner is viewing a teacher's available time slots	the learner selects a slot and clicks "Request Session"	a session request is created with status "Pending" and the teacher receives a notification within 1 minute
a learner has insufficient skill-credits and no cash top-up is configured	the learner attempts to request a paid session	the system blocks the request and prompts the learner to top up credits first

US-E3-02 — Accept / decline session request

Given	When	Then
a teacher has a pending session request	the teacher taps "Accept"	the session status changes to "Confirmed" and both users receive a calendar invite and confirmation notification
a teacher declines a request	the teacher provides an optional reason and taps "Decline"	the session status changes to "Declined", the learner is notified, and any reserved credits are released back to the learner

US-E4-01 — Earn skill-credits after teaching

Given	When	Then
a scheduled session has been marked "Completed" by both participants	the system processes the session closure	the teacher's skill-credit balance is increased by the agreed credit value within 5 minutes

US-E5-01 — Rate and review after session

Given	When	Then
a session has been marked "Completed"	the learner submits a star rating (1–5) and an optional written review within 14 days	the review is published on the teacher's profile and the teacher's average rating is recalculated
a session has been completed	7 days pass without the learner submitting a review	the system sends one reminder notification and then no longer prompts for that session

US-E5-02 — Report inappropriate behavior

Given	When	Then
a user encounters inappropriate content or behavior from another user	the user submits a report with a reason and optional evidence	the report appears in the moderation queue within 1 minute and the reporting user receives a confirmation

4. Product Backlog Prioritization

The product backlog is prioritized using a combination of MoSCoW categorization and a Business Value vs. Effort assessment (Value and Effort scored on a 1–10 scale, where higher value/lower effort ranks higher). This ensures the team delivers maximum customer and business value early, while respecting technical dependencies (e.g., onboarding and profile creation must precede matching and scheduling).

Rank	ID	User Story (short)	MoSCoW	Business Value (1-10)	Effort (1-10)
1	US-E1-01	Sign up / login	Must	9	3
2	US-E1-02	Create profile with skills	Must	10	4
3	US-E2-01	List a skill (teacher)	Must	9	3
4	US-E2-02	Search & filter skills	Must	9	5
5	US-E3-01	Request a session	Must	9	5
6	US-E3-02	Accept/decline session	Must	8	4
7	US-E4-01	Earn skill-credits	Must	8	6
8	US-E5-01	Rate & review	Should	7	4
9	US-E1-03	Identity verification	Should	7	6
10	US-E2-03	View teacher profile/reviews	Should	7	3
11	US-E4-02	Spend credits / cash top-up	Should	7	7
12	US-E3-03	Session calendar view	Should	6	3
13	US-E5-02	Report inappropriate behavior	Should	6	4
14	US-E3-04	Session reminders	Could	5	3
15	US-E1-04	Set location & availability	Should	6	4
16	US-E4-03	Credit balance & history	Could	5	3

Rank	ID	User Story (short)	MoSCoW	Business Value (1-10)	Effort (1-10)
17	US-E5-03	Verification badges	Could	4	2
18	US-E6-01	Push/email notifications	Could	5	4
19	US-E5-04	Moderator dashboard	Could	5	6
20	US-E2-04	Personalized recommendations	Won't (this release)	4	7
21	US-E1-05	Edit/deactivate profile	Could	3	2
22	US-E3-05	Reschedule/cancel session	Could	4	3
23	US-E6-02	Skill groups	Won't (this release)	3	5
24	US-E6-03	Community events	Won't (this release)	3	5

Table 4.1 — Full Product Backlog ranked by MoSCoW priority and Value/Effort. Items ranked 1–7 are selected for Sprint 1.

5. Sprint Backlog (Sprint 1)

Sprint 1 is a 2-week sprint. The seven highest-priority "Must Have" stories (Rank 1–7) are selected, since together they deliver the minimum end-to-end flow: a user can sign up, build a profile, list a skill, discover another user, request and confirm a session, and earn a skill-credit. Each story is broken down into actionable development tasks.

US-E1-01 — Sign up / login

Task	Description	Owner	Est. (hrs)
T1.1	Design sign-up/login UI (email, phone, social login)	UI/UX Designer	6
T1.2	Implement backend auth service (JWT/session, OTP verification)	Backend Dev	12
T1.3	Integrate social login (Google) SDK	Frontend Dev	6
T1.4	Write unit & integration tests for auth flow	QA	5

US-E1-02 — Create profile with skills

Task	Description	Owner	Est. (hrs)
T2.1	Design profile creation/edit screens	UI/UX Designer	6
T2.2	Build "skills to teach / learn" data model & API	Backend Dev	8
T2.3	Implement frontend form with validation	Frontend Dev	6
T2.4	Test profile creation edge cases (empty skills, long text)	QA	4

US-E2-01 — List a skill (teacher)

Task	Description	Owner	Est. (hrs)
T3.1	Design "add skill listing" form (description, level, format)	UI/UX Designer	4
T3.2	Implement skill-listing API & database schema	Backend Dev	8
T3.3	Build frontend listing management screen	Frontend Dev	6
T3.4	Functional testing of listing creation/edit	QA	4

US-E2-02 — Search & filter skills

Task	Description	Owner	Est. (hrs)
T4.1	Design search results & filter UI	UI/UX Designer	5

Task	Description	Owner	Est. (hrs)
T4.2	Implement search/filter API with indexing (category, distance, rating)	Backend Dev	10
T4.3	Integrate frontend search with debounce and pagination	Frontend Dev	8
T4.4	Test filter combinations and empty-result states	QA	5

US-E3-01 — Request a session

Task	Description	Owner	Est. (hrs)
T5.1	Design session request & slot-selection UI	UI/UX Designer	5
T5.2	Implement session request API and status workflow	Backend Dev	10
T5.3	Build notification trigger for new requests	Backend Dev	4
T5.4	Test credit-check and request validation logic	QA	5

US-E3-02 — Accept / decline session

Task	Description	Owner	Est. (hrs)
T6.1	Design accept/decline/propose-time UI for teachers	UI/UX Designer	4
T6.2	Implement status transition API (Pending → Confirmed/Declined)	Backend Dev	8
T6.3	Integrate calendar invite generation on confirmation	Frontend/Backend Dev	6
T6.4	Test decline flow and credit-release logic	QA	4

US-E4-01 — Earn skill-credits after teaching

Task	Description	Owner	Est. (hrs)
T7.1	Design credit ledger data model	Backend Dev	6
T7.2	Implement session-completion trigger to credit teacher	Backend Dev	8
T7.3	Build "mark session complete" UI for both participants	Frontend Dev	5
T7.4	Test credit accuracy and race conditions on dual confirmation	QA	6

6. Story Point Estimation

Story points are estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), reflecting relative complexity, effort, and uncertainty rather than absolute hours. The whole team (Product Owner facilitating, Developers, QA, UI/UX) estimated independently and discussed outliers until consensus was reached.

ID	User Story	Story Points	Justification
US-E1-01	Sign up / login	5	Standard auth pattern, but includes OTP + social login integration, adding moderate complexity and third-party dependency risk.
US-E1-02	Create profile with skills	5	New data model for dual skill lists plus validation logic; UI is moderately complex but no external dependencies.
US-E2-01	List a skill (teacher)	3	Straightforward CRUD feature built on the existing profile/skill data model; low uncertainty.
US-E2-02	Search & filter skills	8	Requires efficient multi-criteria search/indexing (category, geo-distance, rating), pagination, and performance tuning — higher technical complexity.
US-E3-01	Request a session	8	Involves slot selection, credit pre-check, notification triggers, and multiple state transitions — cross-cutting and moderately uncertain.
US-E3-02	Accept / decline session	5	Builds on the request workflow; adds calendar integration, which introduces some external-API uncertainty.
US-E4-01	Earn skill-credits after teaching	5	Requires a reliable ledger and dual-confirmation logic to avoid race conditions; financial-integrity concerns raise complexity.

Sprint 1 Total Story Points: 39

Based on the team's historical/estimated velocity of approximately 35–40 points per 2-week sprint (5-member cross-functional team), this Sprint 1 scope of 39 points is considered achievable and realistic without excessive overtime.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, a new user can register, build a profile listing skills to teach and learn, list a teachable skill, discover another user's skill via search, request and confirm a session with them, and — upon session completion — earn skill-credits, demonstrating a complete end-to-end core exchange loop of the SkillSwap platform."

7.2 Expected Sprint Deliverables

- A functioning sign-up/login flow supporting email, phone OTP, and Google social login.
- A profile creation and editing screen supporting "skills to teach" and "skills to learn" lists.
- A skill-listing feature allowing teachers to publish skill offerings with description, level, and format.
- A search and filter interface enabling learners to discover teachers by category, distance, and rating.
- An end-to-end session request → accept/decline → confirmation workflow with notifications.
- A working skill-credit ledger that credits teachers automatically upon session completion.
- Automated unit/integration test coverage for all Sprint 1 features and a demo-ready build for Sprint Review.

7.3 Definition of Done

- Code is peer-reviewed, merged to the main branch, and passes CI build.
- Unit and integration tests are written and passing (minimum 80% coverage for new modules).
- Acceptance criteria for the story are verified and signed off by QA.
- UI matches approved design mock-ups and passes basic accessibility checks.
- Feature is deployed to the staging environment and demonstrated at Sprint Review.
- No critical or high-severity bugs remain open against the story.

7.4 Summary

This sprint plan translates the SkillSwap problem statement into a prioritized, estimated, and actionable Sprint 1 backlog. By focusing the first sprint on the core registration → listing → discovery → session → credit-exchange loop, the team validates the platform's central value proposition early, enabling faster feedback and informed planning for subsequent sprints covering trust & safety, payments, and community features.